# Offline Data Preprocessing Pipeline

*Analysis of preprocessing.py*

Automated Research Report

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This document provides a detailed breakdown of the offline preprocessin
pipeline found in preprocessing/preprocessing.py. This pipeline standar
DICOM/NIfTI scans across multiple datasets (BHSD, CQ500, PhysioNet) int
clean, uniform format ready for 3D Segmentation Models.
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## Phase 1: Spatial & Geometric Standardization

- ### Orthonormalize Direction Cosines

  Why: Some CT scanners save images with slightly skewed transformation matri
  mathematically corrects the affine matrix to be perfectly orthogonal (90-de
  angles). This is absolutely critical for 3D CNNs, otherwise the convolution
  distort the physical anatomy.

- ### Isotropic Resampling

  Why: CT scans from different hospitals have different voxel spacings (e.g.,
  X/Y, but 5.0mm in Z). This resamples every patient to a standard physical r
  (e.g., 1x1x1 mm) using Linear interpolation for images and Nearest Neighbor
  masks, ensuring a 3x3x3 CNN kernel always represents the exact same physica
  across all patients.

- ### Z-Axis Depth Standardization (standardize_depth)

  Why: Scans have a different number of slices (e.g., 30 vs 150). This pads o
  the Z-axis to a fixed depth (e.g., 128 slices). Crucially, if it crops, it
  the centroid of the hemorrhage and ensures the lesion is perfectly centered
  crop, preventing the ground truth from being cut off.

## Phase 2: Intensity & Contrast Normalization

- **CT Windowing (apply_window)**

  Why: Raw Hounsfield Units (HU) range from -1000 to +3000, but blood is only
  in a very narrow band. This applies a 'Brain Window' (or multi-channel wind
  brain, bone, and subdural) to clamp intensities, drastically boosting the c
  the hemorrhage against gray/white matter.

- **Histogram Matching**

  Why: Different scanner manufacturers (GE, Siemens, Philips) have different
  intensity profiles. This forces the intensity histogram of every scan to ma
  'golden reference' scan, completely eliminating scanner-specific intensity

- **Z-Score / Min-Max Normalization**

  Why: Neural networks struggle with large numbers. This normalizes the windo
  intensities to either a strict [0, 1] range or a Gaussian distribution (mea
  std=1), optimizing gradient flow during backpropagation.

## Phase 3: Label Cleaning & Quality Control

- **Mask Morphological Cleaning (clean_mask)**

  Why: Removes tiny, isolated false-positive pixels in the manual annotations
  Connected Component Analysis. It also uses BinaryFillhole to fill any accid
  inside a hemorrhage annotation, ensuring the CNN learns solid, continuous l
  structures.

- **Automated Quality Control (QC) Generation**

  Why: Automatically generates a 3-panel visualization (middle slice, max-pat
  slice with red overlay, and binary mask) for every processed patient. This
  researchers to rapidly scan hundreds of patients visually without loading m
  NIfTI files into viewers like 3D Slicer.

- **Reproducibility Hashing (build_metadata)**

  Why: Embeds the current Git commit, configuration SHA256 hash, and Python/L
  versions directly into the JSON metadata of every processed scan. This ensu
  traceability for FDA/CE medical device compliance.